

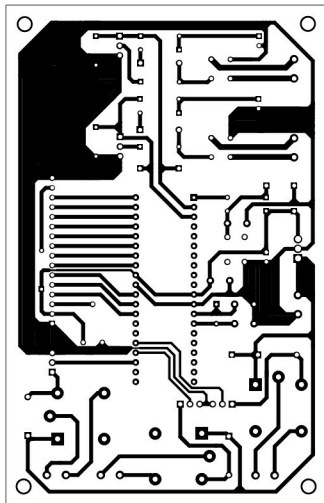


Fig. 3: An actual-size, single-side PCB for the PFC

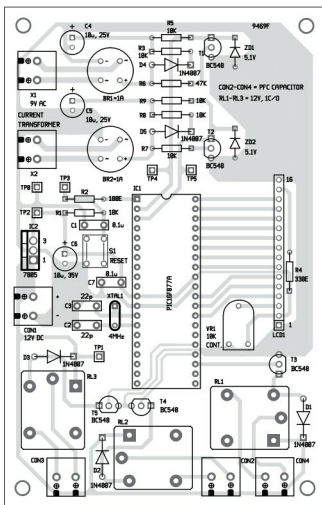


Fig. 4: Component layout for the PCB

connected across the load. There are various methods to determine the corrective (leading) reactive power (kVAR) rating. Here we present a very simple method to know the kVAR in two steps:

1. Measure the RMS line voltage (line-to-neutral for single-phase load) and RMS line current. Power factor is already displaying on the LCD. For a single-phase system:

$$\text{kW} = \text{volts} \times \text{amps} \times \text{PF} / 1000$$

2. Now determine the corrective (leading) reactive power (kVAR) needed for obtaining the desired PF by multiplying the kW value, as found above, with the selected value of power factor multiplier from Table I (intersection point of original PF in column 1 and the desired PF). For example, if the total plant load is 100kW at 60 per cent power factor. Capacitor

kVAR rating necessary to improve power factor to 90 per cent is found by multiplying kW (100) by the 0.849 multiplier in the table, which gives 84.9kVAR, or roughly 85kVAR. The standard PF capacitor rating nearest to 85kVAR should be used.

Software

The program is written in 'C' language and compiled using HI-TECH compiler along with MPLAB to generate hex code. The generated hex code is burnt into the microcontroller using suitable programmer with configuration bit setting as shown in Fig. 2. Program is well commented and easy to understand.

When the system is switched on, the main program initiates the LCD. Depending on the power factor across the load the 'PF is too low' message will be displayed in the first line and

power factor value, say, PF=56.70 per cent, in the second line of the LCD. Timer starts when a positive-going pulse is received at port pin RA0 and stops when port pin RA1 receives positive-going pulse. This timer count is converted to radians. Cosine value of the radians is the original power factor displayed on the LCD. One of the Port C pins (RC4, RC5 or RC6—depending on the power factor) goes high and energises the corresponding relay to connect the respective PFC capacitor across the load.

Construction and testing

An actual-size, single-side PCB for the PFC circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the recommended PCB to save time and minimize assembly errors. Carefully assemble the com-